Department of Artificial Intelligence and Machine Learning

Date: 16/04/2026	**Test – I**	**Max. Marks: 50**
Semester: VI	**UG**	**Duration: 1.5 Hours**
Course Title: Cloud Computing Technology and Architectures		**Course Code: AI364TA**

Q. No	Question	M	BT	CO
1.a	Explain the Cloud Computing Reference Model and describe the roles of: • Infrastructure as a Service (IaaS) • Platform as a Service (PaaS) • Software as a Service (SaaS) Provide one example for each layer.	5	2	1
1.b	A company plans to deploy its services on a cloud-based infrastructure. While implementing the system, it encounters several technical and operational issues. For each of the following scenarios, identify the cloud computing challenge involved. 1. A financial institution wants to store sensitive customer transaction data on a cloud platform but is concerned about unauthorized access and data breaches. 2. A cloud-based e-commerce platform experiences sudden spikes in user traffic during festive sales, leading to delays in processing customer requests. 3. An organization uses cloud services from different vendors but finds it difficult to integrate and migrate applications across these platforms. 4. A healthcare company stores patient data on cloud servers located in different countries and must comply with varying data protection regulations. 5. A cloud-based video streaming platform must ensure that its services remain available even if one or more data center servers fail.	5	3	1
2.a	Consider the following scenarios. Identify and briefly explain the cloud computing characteristic being demonstrated in each case. 1. A video streaming company automatically increases the number of servers during major sporting events when millions of users access the service simultaneously. 2. A startup rents virtual machines from a cloud provider and pays only for the hours the machines are used. 3. Employees of a multinational company access enterprise applications from laptops, tablets, and smartphones through the internet.	6	3	1
2.b	Distinguish between Shared Memory and Distributed Memory MIMD Architectures.	4	2	1
3.a	Identify the level of parallelism demonstrated in the following scenarios: 1. Multiple arithmetic operations within a processor instruction are executed simultaneously. 2. A multi-core processor executes several threads of the same program concurrently. 3. A scientific simulation is divided into multiple independent tasks that run on separate machines.	6	3	1
3.b	Explain the following architectural styles used in distributed computing, highlighting their structure and typical use cases: • Blackboard • Pipe and Filter	4	2	1

4.a	A cloud data center hosts applications from several organizations on the same physical infrastructure using virtual machines. Based on this scenario, explain how virtualization provides the following characteristics: • Isolation • Portability • Managed execution • Resource sharing	5	3	2
4.b	Explain the concept of hypervisors and differentiate between Type I and Type II hypervisors with suitable examples.	5	2	2
5.a	Explain the taxonomy of virtualization techniques, describing the working principles of: • Full virtualization • Paravirtualization • Partial virtualization	5	2	2
5.b	Compare Xen and Microsoft Hyper-V virtualization technologies with respect to: • Virtualization approach • Hypervisor type • Typical use cases	5	4	2

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Date: 30/05/2026	**Test – II**	**Max. Marks: 50**
Semester: VI	**UG**	**Duration: 90 minutes**
Course Title: Cloud Computing Technology and Architectures		**Course Code:AI364TA**

Q. No.	Question	M	BT	CO
1.a	Explain the Cloud Reference Model with a neat diagram. Discuss the roles of the major entities involved in cloud service delivery.	6	2	1
1.b	Differentiate between the Public, Private, Hybrid and Community cloud deployment models with respect to how they work and one suitable example for each.	4	2	1
2.a	A video streaming company receives millions of live user interactions, search queries, video uploads, and recommendation requests every minute from users across the world. Analyze how Data-Intensive Computing can be used to handle such large-scale streaming and distributed data efficiently. Discuss the major characteristics and challenges involved in processing this data.	6	3	4
2.b	Classify the following scenarios as Compute-Intensive, Data-Intensive, or Both. Justify your answer briefly: 1) Real-time weather forecasting using supercomputers 2) Social media platforms storing and analyzing petabytes of user posts and images 3) AI-based video surveillance system performing live facial recognition on streaming CCTV data 4) Scientific simulation of galaxy formation using high-performance computing clusters	4	4	4
3.a	Explain the working of the MapReduce programming model using the Word Count problem. Illustrate the Map phase and Reduce phase with suitable examples.	6	3	4
3.b	A large-scale e-commerce company requires a distributed storage system capable of handling millions of daily user requests with high availability and scalability across geographically distributed datacenters. In this context: 1. Explain how Amazon Dynamo achieves scalability and high availability. 2. Discuss the role of Google File System (GFS) in supporting large-scale distributed storage.	4	3	4
4.a	Explain the platform architecture of Google App Engine with a neat diagram.	5	2	2
4.b	Discuss the Fabric Controller and the role-based architecture of Microsoft Azure.	5	2	2
5	A multinational healthcare organization plans to adopt a hybrid multi-cloud strategy to manage patient records, medical imaging data, AI-based diagnosis systems, and telemedicine services. The organization wants to ensure regulatory compliance, scalability, interoperability, and secure access across multiple cloud providers. As a cloud architect, explain how the TOGAF ADM cycle can be used to design the organization's enterprise cloud strategy. Discuss the role of the BDAT architecture domains during cloud adoption while explaining how these domains help in planning, integration, governance, and implementation of hybrid and multi-cloud environments.	10	4	3

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Academic Year: 2025-2026 Even

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Date: 18/06/2026	Test – III	Max. Marks: 50
Semester: VI	UG	Duration: 1.5 Hours
Course Title: Cloud Computing Technology and Architectures		Course Code: AI364TA

Q. No	Question	M	BT	CO
1.a	Describe the role of Active Directory as a central identity store in a multi-cloud environment. Discuss how federation helps in managing user identities across multiple cloud platforms.	6	2	5
1.b	Differentiate between Authentication and Authorization in IAM.	4	2	5
2.a	Discuss the importance of architecture principles in designing multi-cloud solutions. Discuss any four quality attributes that should be considered while designing an enterprise cloud architecture.	6	2	4
2.b	For each of the following scenarios, identify the most appropriate architecture principle category and justify your answer: a. A healthcare organization stores patient records in a multi-cloud environment. The architecture team must ensure compliance with data privacy regulations and implement strict access controls to prevent unauthorized access. b. An e-commerce company expects traffic to increase fivefold during festive sales. The cloud architecture must support auto-scaling, high availability, and deployment across multiple cloud regions.	4	3	4
3.a	Outline the steps involved in implementing DevOps in a multi-cloud environment.	5	3	5
3.b	Draw and illustrate the Build and Release Pipeline used in CI/CD. Clearly indicate the stages from code commit to deployment.	5	3	5
4	A software development team working on a banking application directly pushes code changes to the main branch without using pull requests, code reviews, or testing pipelines. During a release, a faulty code commit causes a service outage affecting customers. a. Analyze the consequences of pushing code directly to the main branch. b. Recommend best practices that should be followed using CI/CD and version control mechanisms to prevent such incidents in future.	10	4	5
5.a	A multinational organization operates applications across AWS, Azure, and Google Cloud. The operations team is experiencing frequent performance degradation, delayed incident detection, and inefficient resource utilization. Using the components and layered architecture of AIOps, demonstrate how AIOps can be applied to improve monitoring, incident management, and optimization in this multi-cloud environment.	6	3	4
5.b	A company operating application across multiple cloud providers wants to improve observability and operational monitoring. Explain how Prometheus and Grafana can be used together to support monitoring, analytics, and visualization in this environment.	4	3	4